

CS-540 Data Literacy & Visualizations

**6-2 Submit Activity: Advanced Data Visualization with Python**

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## **Reflection: Visualization Design, Interactivity, and Data Modeling**

### **Design Choices and Their Impact on Clarity**

The purpose of this project was to design and implement a dynamic data visualization dashboard using Python to explore COVID-19 mortality data across countries and time. Several visualization techniques were implemented using Matplotlib, Seaborn, and Plotly to present the data in a clear and meaningful way. The primary goal of the design process was to transform a large and complex dataset into visualizations that communicate insights effectively while staying accessible to a wide audience.

One of the first design decisions involved preparing the dataset so that it could be used effectively for visualization. The dataset held daily COVID-19 death counts for many countries, along with aggregated regions such as “World” and continental groups. The preprocessing stage involved cleaning the dataset to remove rows and columns with missing or zero values and converting the date column into a proper datetime format. This step ensured that time-based visualizations would display correctly and that the data used for analysis reflected meaningful trends rather than empty or incomplete entries. Also, countries with no recorded deaths were removed so that the visualizations would focus on relevant data and reduce unnecessary clutter. Another key preparation step involved selecting the countries that would be analyzed in detail. Instead of displaying every country in the dataset, the notebook found the countries with the highest total deaths. This allowed the visualizations to focus on the most significant cases while keeping the charts readable. Limiting the number of countries displayed reduced cognitive overload and ensured that viewers could easily compare trends across major countries without being overwhelmed by hundreds of lines or bars. Several static visualizations were created using

Matplotlib and Seaborn to highlight key insights from the dataset. The first static visualization was a correlation heatmap, which showed the relationship between death trends in the top countries. Heatmaps are effective when analyzing correlations because they allow viewers to visually compare relationships between multiple variables at once. In this visualization, the color gradient helped show the strength of correlations between different countries' death patterns. By examining the heatmap, viewers could quickly see which countries had similar pandemic trends over time. The second static visualization was a bar chart comparing total deaths among the top twenty countries. Bar charts are useful for comparing categorical data because they clearly show differences in size between categories. A horizontal bar chart format was chosen to make the country's names easier to read and to improve the overall visual balance of the chart. Sorting the bars by total deaths further enhanced clarity, allowing viewers to at once find which countries were most affected. The third static visualization was a time series line chart showing the progression of total deaths over time. Time series charts are essential when analyzing changes across time periods. In this visualization, the global death count was plotted across dates to show how the pandemic evolved. Labels, titles, and axis formatting were carefully added to ensure that viewers could easily interpret the chart. Rotating the date labels and adjusting the chart dimensions improved readability, particularly when displaying long time ranges.

## **Interactive Visualizations**

In addition to static charts, interactive visualizations were created using Plotly to allow users to explore the data more deeply. Interactive visualizations provide advantages over static charts because users can zoom, filter, and inspect specific values in real time. The first interactive visualization implemented was a drill-down or zoomable chart showing global death trends over

time. Using Plotly's interactive features, users can zoom into specific date ranges to examine phases of the pandemic more closely. Tooltips were also included so that hovering over a point on the chart displays the exact value and date associated with that point. These interactive elements allow viewers to investigate the data without requiring more charts. The second interactive visualization implemented the overview-plus-detail design pattern. This visualization displayed two coordinated charts: an overview chart showing general trends and a detailed chart showing more granular information. The overview allowed users to show the general pattern across the dataset, while the detailed chart allowed them to focus on specific countries and time periods. This design is particularly effective for large datasets because it allows users to keep context while exploring specific details. Interactive elements such as tooltips, zoom functionality, and filtering capabilities improved the usability of the dashboard. These features help users explore the dataset dynamically instead of relying solely on static representations.

### **Customization and Visual Theme**

Customization was applied throughout the project to ensure a consistent visual theme and improve usability. This included setting up consistent fonts, color palettes, and layout formatting across the different visualizations. Gridlines were used selectively to improve readability without overwhelming the charts. Color palettes were chosen carefully so that different data elements could be distinguished easily. For example, diverging color scales were used in the heatmap to emphasize differences in correlation strength, while consistent line colors were used for country trends in the time series visualizations. The layout of each visualization was also customized for clarity. Titles were descriptive and clearly explained what each chart represented. Axis labels were included so that viewers could interpret values accurately. Legends were added where

multiple categories were displayed to ensure that each visual element could be named easily. These design choices ensured that the dashboard remained visually coherent even though it included multiple visualization types.

### **Animation and Transition Effects**

One of the most advanced components of the project was the implementation of animation using Matplotlib. The animation visualized how COVID-19 death trends changed over time for the selected countries. This required reshaping the dataset so that each country could be plotted dynamically across periods. The animation was implemented using Matplotlib's FuncAnimation function. This required several steps, including creating line objects for each country, defining an initialization function to reset the lines, and defining an update function that updated the chart for each frame. The animation was then made inside the notebook using HTML so that it could be exported and displayed in the final HTML report. Animating the data added an added dimension to the visualization by allowing viewers to see how trends evolved over time rather than viewing static snapshots. This helped illustrate the progression of the pandemic in a more engaging and intuitive way.

### **Challenges and Solutions**

Several challenges were met while implementing the interactive and animated components of the project. One of the most common issues involved mismatches between the dataset structure and the expectations of certain visualization functions. For example, some functions expected the dataset to hold columns named "location" and "total\_deaths," while the original dataset stored countries as separate columns. This caused errors when trying to perform grouping operations.

To address this issue, the dataset was reshaped when necessary, using data transformation techniques such as selecting relevant columns and using the melt function to convert the dataset into a format suitable for animation. Understanding how data structure affects visualization tools was an important lesson learned during this process. Another challenge involved the execution of order of cells within the Jupyter Notebook environment. Because variables are only available after the cell that defines them has been executed, running cells out of order caused several `NameError` exceptions. This issue was resolved by restarting the kernel and running the entire notebook sequentially from top to bottom to ensure that all variables were defined before being used. Rendering the animation inside the notebook also presented difficulties. Initially, the animation did not display correctly because the notebook environment required the animation to be converted to an HTML format. This problem was solved by using the `to_jshtml()` method and displaying the result using the IPython display tools.

### **Accessibility and Cognitive Load**

Accessibility and cognitive load were important considerations throughout the design process. Because the dataset has many countries and time points, displaying too much information simultaneously could make the visualizations difficult to interpret. To reduce cognitive load, the analysis focused on the most relevant countries rather than displaying all countries in the dataset. This approach made comparisons easier and prevented charts from becoming overcrowded. Color choices were also selected with accessibility in mind. High-contrast color palettes were used so that viewers could easily distinguish between different values and categories. Clear labeling and descriptive titles ensure that users could interpret the charts without needing another context. Interactive features also helped reduce cognitive load by allowing users to explore the

data gradually rather than viewing all information at once. For example, zooming and hover tooltips allow users to focus on specific data points when needed.

In conclusion, this project showed how thoughtful data preparation, visualization design, and interactive features can transform complex datasets into clear and meaningful insights. Static visualizations provided an overview of key trends, while interactive charts allowed users to explore the data more deeply. Animation further enhanced visualization by illustrating how trends evolved over time. The challenges met during development reinforced the importance of understanding both data structure and the technical requirements of visualization libraries. By addressing these challenges and applying consistent design principles, the final dashboard effectively communicates the progression and impact of COVID-19 across different countries and time periods.